squeeze (Chapter 26). That was our highly educated guess. We were wrong.

In the intervening quarter century, two additional singularities were discovered, mathematically, inside black holes: *gentle* singularities, to the extent that any singularity can be gentle (Chapter 26). Gentle enough that Cooper, falling into one, might possibly survive. I'm dubious of survival, but we can't be sure. So I now think it respectable, in science fiction, to posit survival.

Also in the intervening quarter century, we have learned that our universe is probably a brane in a higher-dimensional bulk (Chapter 21). So it's respectable, I think, to posit living beings that inhabit the bulk—a very advanced civilization of bulk beings—who might save Cooper from the singularity at the last moment. That's what Christopher Nolan chose.

Through the Event Horizon



In *Interstellar*, when Ranger 2 piloted by Cooper (and lander 1, piloted by TARS) eject from the *Endurance*, they spiral down toward Gargantua's event horizon and then through it. What do Einstein's relativistic laws say about this downward spiral?

According to those laws, and hence my interpretation of the movie, Brand, watching from the *Endurance*, can never see the Ranger penetrate the horizon. No signal Cooper tries to send her from inside the horizon can ever get out. The flow of time inside the horizon is downward, and that downward time flow drags Cooper and all signals he sends downward with itself, away from the horizon. See Chapter 5.

So what *does* Brand see (if she and Case can stabilize the *Endurance* long enough for her to watch)? Because the *Endurance* and